Arithmetic & Number Sense

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

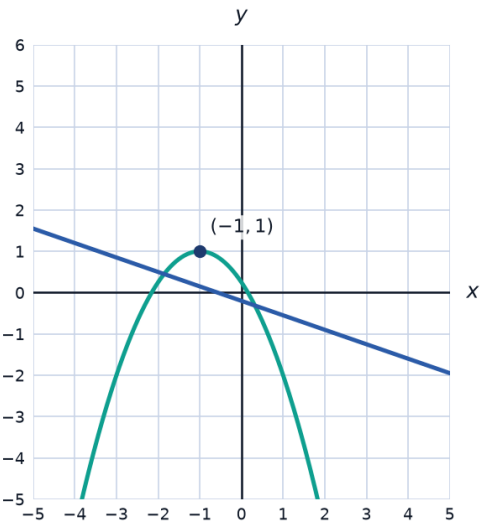
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Order of operations

Evaluate $12 + 6 \times (9 - 5) \div 3$.

2 [Foundational] • Target: Order of operations

Evaluate $8 + 2 \times 5^2 - 30 \div 6$.



3 [Foundational] • Target: Order of operations

Evaluate $\frac{18 - 3 \times 4}{2} + 7$.

4 [Foundational] • Target: Order of operations

Evaluate $(7 - 2)^2 - 3 \times (6 - 4)$.

5 [Foundational] • Target: Powers and roots

Evaluate 3^4.

6 [Foundational] • Target: Powers and roots

Evaluate $\sqrt{144}$.

7 [Foundational] • Target: Powers and roots

Evaluate $2^3 \times 2^2$ using the laws of exponents.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Powers and roots

Evaluate $\sqrt{81} + \sqrt{16}$.

9 [Intermediate] • Target: Working with fractions

Compute $\frac{3}{4} + \frac{5}{6}$.

10 [Intermediate] • Target: Working with fractions

Compute $\frac{7}{8} - \frac{1}{3}$.

11 [Intermediate] • Target: Working with fractions

Compute $\frac{2}{3} \times \frac{9}{10}$.

12 [Intermediate] • Target: Working with fractions

Compute $\frac{5}{6} \div \frac{5}{9}$.

13 [Intermediate] • Target: Working with fractions

Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?

14 [Intermediate] • Target: Decimals

Compute $3.45 + 2.7$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Decimals • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Compute $8.6 - 3.25$.

16 [Challenging] • Target: Decimals • Cross-Lesson: 3.01 Algebra Essentials

Compute 1.2×0.5.

17 [Challenging] • Target: Decimals • Cross-Lesson: 4.01 Geometry & Measurement

Compute $7.5 \div 0.25$.

18 [Challenging] • Target: Place value and order of magnitude • Cross-Lesson: 5.01 Quantitative Comparisons

Write $4{,}500{,}000$ in the form $a \times 10^n$ with $1 \le a < 10$.

19 [Challenging] • Target: Place value and order of magnitude • Cross-Lesson: 6.01 Sequences & Patterns

Which is greater, 6×10^4 or $58{,}000$?

20 [Challenging] • Target: Place value and order of magnitude • Cross-Lesson: 7.01 Data Analysis & Probability

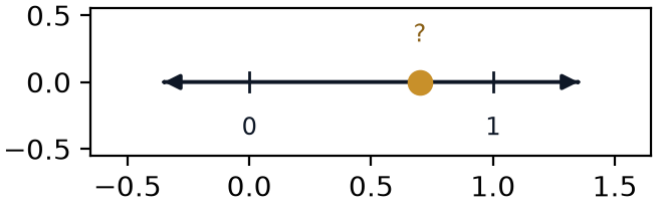
Order from least to greatest: 0.6, $\frac{2}{3}$, 0.65.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Comparing fractions and decimals

Which value is largest: the point shown on the number line, $\frac{3}{5}$, or 0.55?



a. The marked point (0.7)

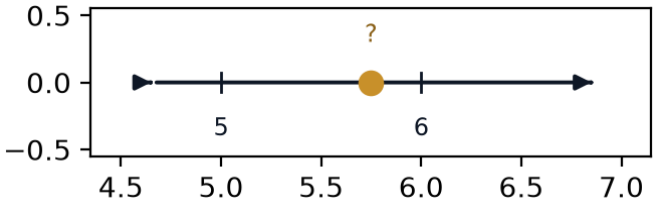
b. $\frac{3}{5}$

c. They are all equal

d. 0.55

22 [Foundational] • Target: Decimal operations

The number line marks the value of $9.2 - 3.45$. Where does it land?



a. 5.75

b. 6.25

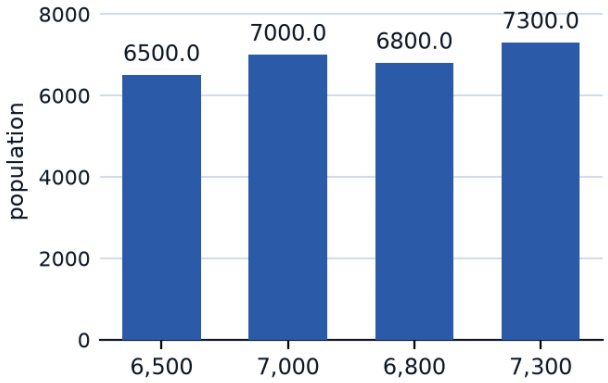
c. 5.85

d. 5.65

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23 [Foundational] • Target: Scientific notation & place value

The bar chart shows four city populations. Which one equals 7×10^3?



a. 7,000

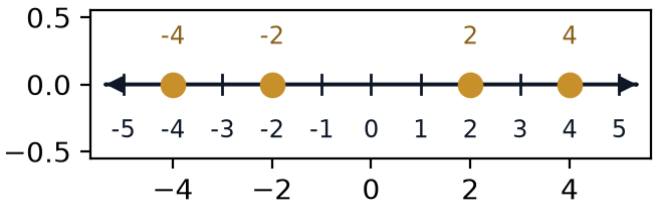
b. 6,800

c. 6,500

d. 7,300

24 [Foundational] • Target: Absolute value & signed numbers

Two of the four points on the number line represent a pair of numbers that are equal in absolute value but opposite in sign. Which pair?



a. −2 and 2

b. −4 and 4

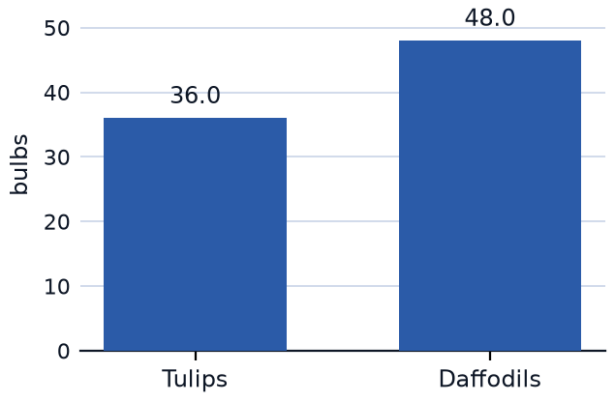
c. −2 and 4

d. −4 and 2

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25 [Foundational] • Target: GCD, LCM & factors

The bar chart shows a gardener's bulb counts. They want to plant them in identical rows with the largest possible number of bulbs per row, each row containing only one type of flower. How many bulbs per row?



a. 4

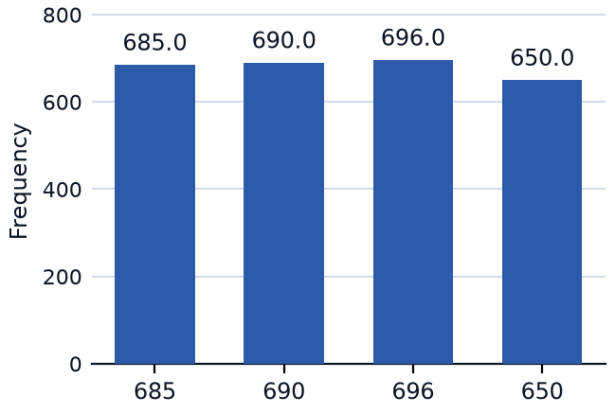
b. 24

c. 6

d. 12

26 [Foundational] • Target: Divisibility rules

The bar chart shows four candidate numbers. Which one is divisible by both 5 and 6?



a. 690

b. 650

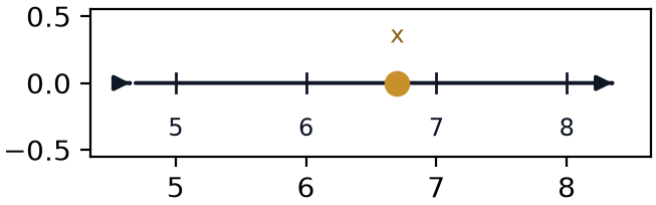
c. 685

d. 696

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27 [Foundational] • Target: Estimation & rounding

The number line shows the value x rounded to the nearest whole number. If $x = 6.7$, where does the rounded value land?



a. 6

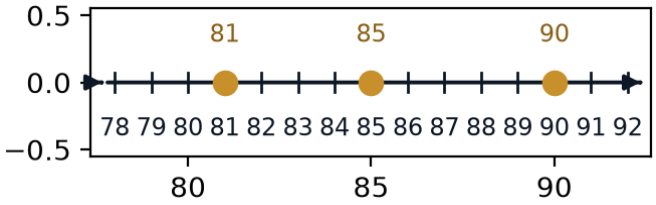
b. 6.5

c. 8

d. 7

28 [Foundational] • Target: Number properties

The number line marks 85 between consecutive multiples of 9. Find the remainder when 85 is divided by 9.



a. 3

b. 4

c. 5

d. 76

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Order of operations

Evaluate $20 - 4 \times 3 + 6 \div 2$.

a. 11

b. 27

c. 9

d. 17

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30 [Intermediate] • Target: Order of operations

Evaluate $(5 + 3)^2 - 2 \times 7$.

a. 78

b. 36

c. 50

d. 46

31 [Intermediate] • Target: Order of operations

Evaluate $\frac{45}{3 + 6} \times 2$.

a. 10

b. 90

c. 7

d. 15

32 [Intermediate] • Target: Order of operations

Evaluate $3 + 2 \times (8 - 5)^2$.

a. 21

b. 45

c. 17

d. 33

33 [Intermediate] • Target: Order of operations

Evaluate $100 - 3 \times (4 + 2 \times 3)$.

a. 64

b. 91

c. 88

d. 70

34 [Intermediate] • Target: Exponents and roots

Use the laws of exponents to evaluate $3^2 \times 3^4$.

a. 243

b. 81

c. 729

d. 27

35 [Intermediate] • Target: Exponents and roots

Evaluate $5^3 - 4^2$.

a. 109

b. 121

c. 93

d. 45

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36 [Intermediate] • Target: Fraction operations

Compute $\frac{1}{3} + \frac{1}{4} + \frac{1}{6}$.

a. $\frac{1}{2}$

b. $\frac{2}{3}$

c. $\frac{3}{13}$

d. $\frac{3}{4}$

37 [Intermediate] • Target: Scientific notation & place value

Evaluate 3.4×10^4 as an ordinary number.

a. 3,400,000

b. 3,400

c. 340,000

d. 34,000

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Scientific notation & place value • Cross-Lesson: 5.01 Quantitative Comparisons

In the number $528{,}437$, what is the place value of the digit 8?

a. Hundred thousands

b. Thousands

c. Ten thousands

d. Hundreds

39 [Challenging] • Target: GCD, LCM & factors • Cross-Lesson: 6.01 Sequences & Patterns

Find the GCD of 54 and 72.

a. 9

b. 6

c. 18

d. 36

40 [Challenging] • Target: GCD, LCM & factors • Cross-Lesson: 7.01 Data Analysis & Probability

Find the LCM of 9 and 15.

a. 3

b. 24

c. 45

d. 135

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41 [Challenging] • Target: GCD, LCM & factors • Cross-Lesson: 8.01 Word Problems & Reasoning

How many factors does 60 have?

a. 6

b. 8

c. 12

d. 10

42 [Challenging] • Target: GCD, LCM & factors • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Is 119 a prime number?

a. No — $119 = 7 \times 17$

b. Yes, it is prime

c. No — $119 = 11 \times 9$

d. Cannot be determined

43 [Challenging] • Target: Divisibility rules • Cross-Lesson: 3.01 Algebra Essentials

Without dividing, is $5{,}481$ divisible by 3?

a. Cannot tell without dividing

b. Yes

c. No

d. Only by 9, not 3

44 [Challenging] • Target: Estimation & rounding • Cross-Lesson: 4.01 Geometry & Measurement

Round 147.362 to the nearest tenth.

a. 147.3

b. 150

c. 147.36

d. 147.4

45 [Challenging] • Target: Estimation & rounding • Cross-Lesson: 5.01 Quantitative Comparisons

Estimate 312×48 by rounding each factor to the nearest ten, then compare to the exact product $14{,}976$.

a. Estimate $310 \times 50 = 15{,}500$, far from the exact value

b. Estimate $300 \times 50 = 15{,}000$, close to the exact value

c. Estimate $320 \times 40 = 12{,}800$, close to the exact value

d. Estimate $310 \times 50 = 15{,}500$, close to the exact value

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